SIMATS ENGINEERING
ASSESSMENT TOOL II – DATA ANALYSIS PROJECT

Course: Data Handling and Visualization	**Code: DSA0601**	**CO:** CO3
Duration: 120 min	**Total Marks:** 100	Reg No : 192424238 Name: Jayaprakash Nivas

COURSE OUTCOME COVERED

CO3: Analyze time-series data, trends, associations, and uncertainty through appropriate visualization methods. (BL4)

Activity Tool : Data Analysis Project

Weightage: 20%

Code of Conduct:

I Jayaprakash Nivas [192424238] (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Problem Understanding & Data Preparation 25%	Data Analysis & Interpretation 30%	Application of Analytical Techniques 20%	Analysis of Results 15%	Organization & Documentation 10%	Presentation & Clarity 5%	Total
Q1							
Q2							
Q3							
Q4							
Q5							

Data Analysis Project

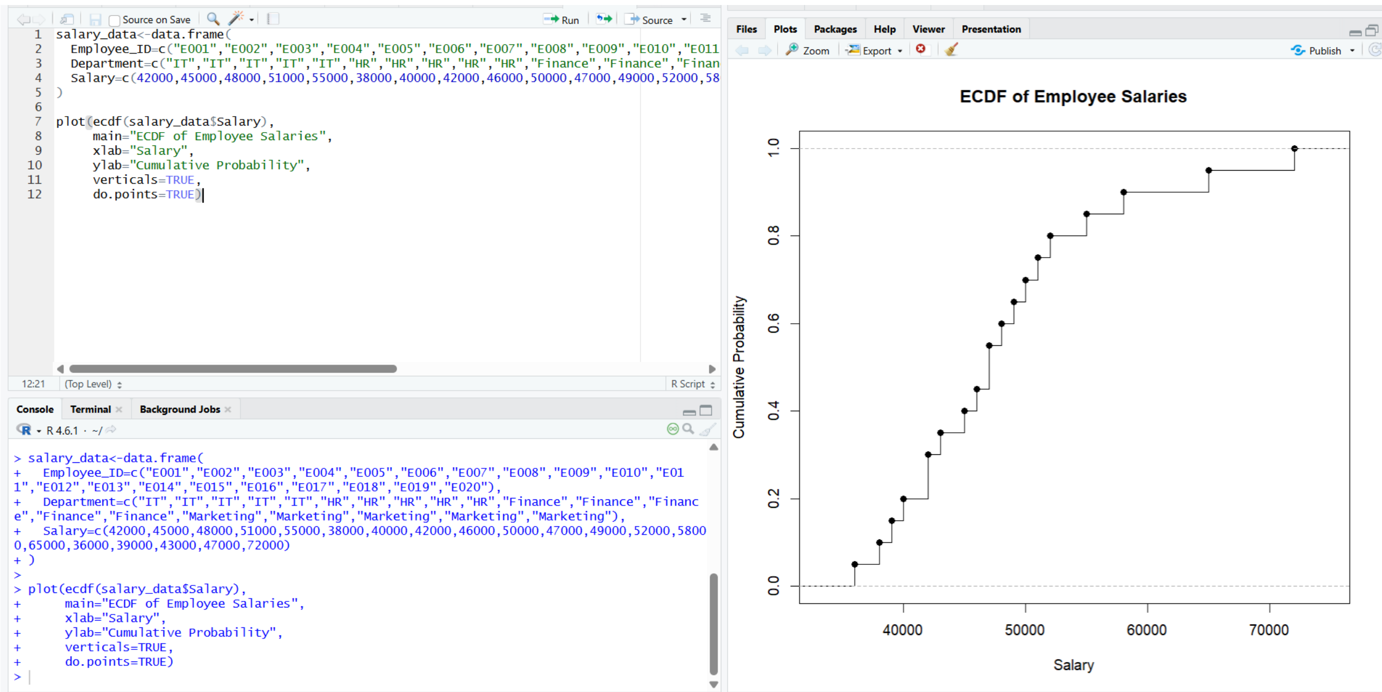
Question 1 – Distribution Analysis Using ECDF and Q-Q Plot

Dataset: Employee_Salaries.csv

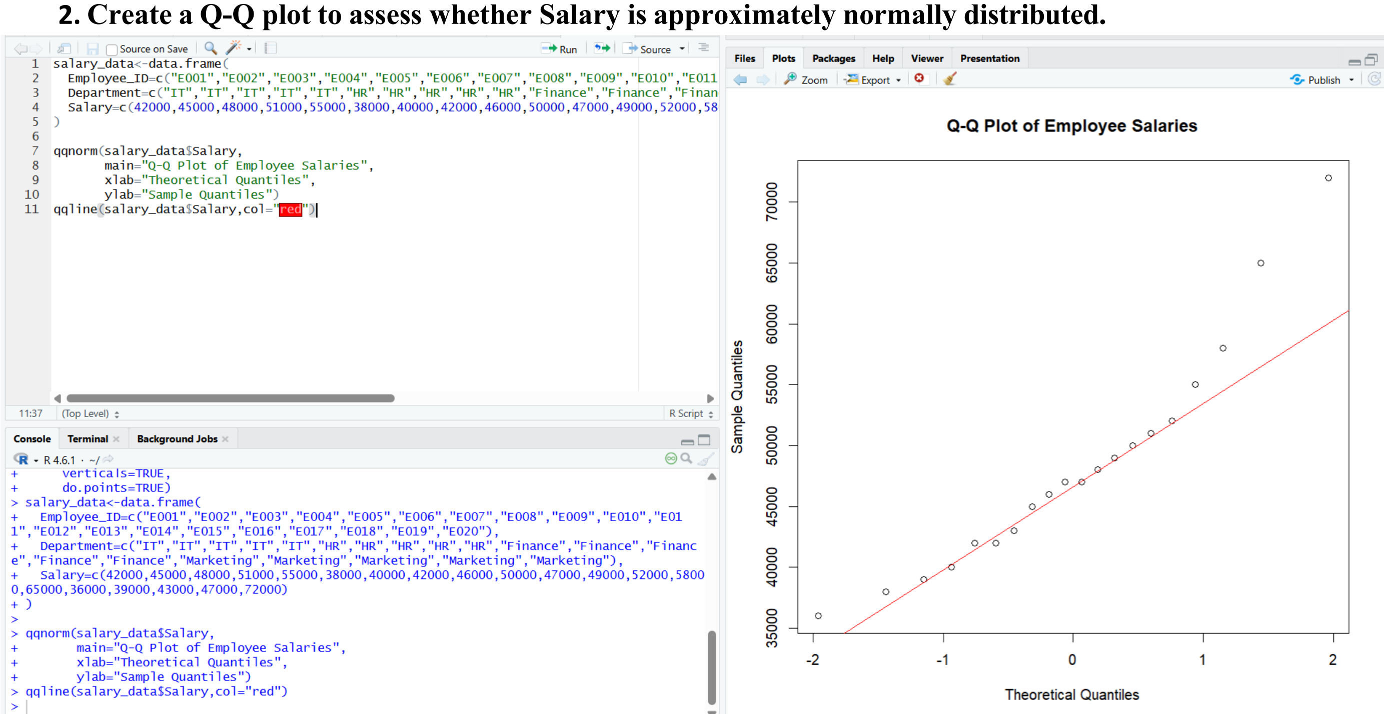
Assessment Task

Using the Employee_Salaries dataset, analyze the distribution of employee salaries.

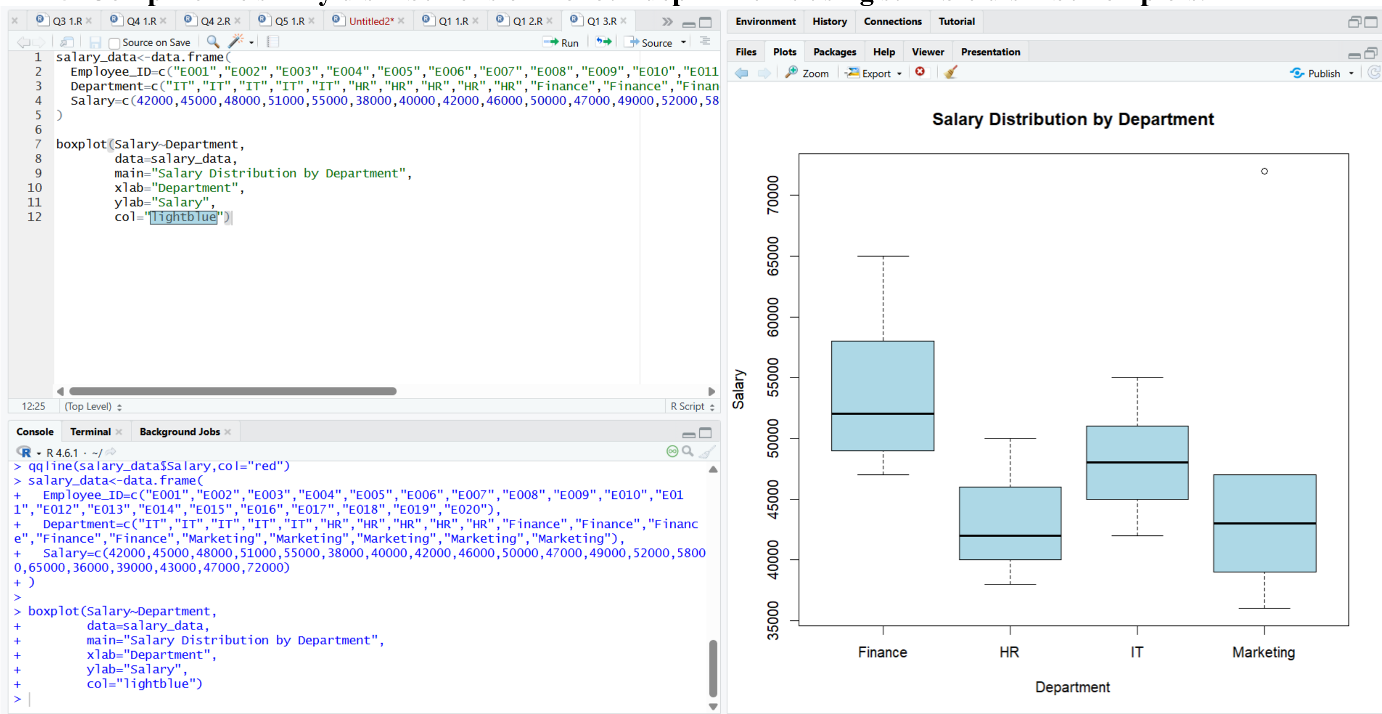
1. Create an empirical cumulative distribution function (ECDF) for Salary.



2. Create a Q-Q plot to assess whether Salary is approximately normally distributed.



3. Compare the salary distributions of the four departments using suitable distribution plots.



4. Identify any possible skewness or unusual observations and explain what the plots reveal.

The salary distribution shows right skewness, mainly because of the high salary values in the upper range. The $72,000 salary of the Marketing department is a particularly unusual observation compared with the other salaries. The ECDF shows a longer upper tail, while the Q-Q plot shows deviation from the reference line at the extreme values, indicating that the salary distribution is not perfectly normal.

Dataset:

Employee_ID	Department	Salary
E001	IT	42000
E002	IT	45000
E003	IT	48000
E004	IT	51000
E005	IT	55000
E006	HR	38000
E007	HR	40000
E008	HR	42000
E009	HR	46000
E010	HR	50000
E011	Finance	47000
E012	Finance	49000
E013	Finance	52000
E014	Finance	58000
E015	Finance	65000
E016	Marketing	36000
E017	Marketing	39000
E018	Marketing	43000

E019	Marketing	47000
E020	Marketing	72000

Question 2 – Visualizing Many Distributions Using Bean Plots

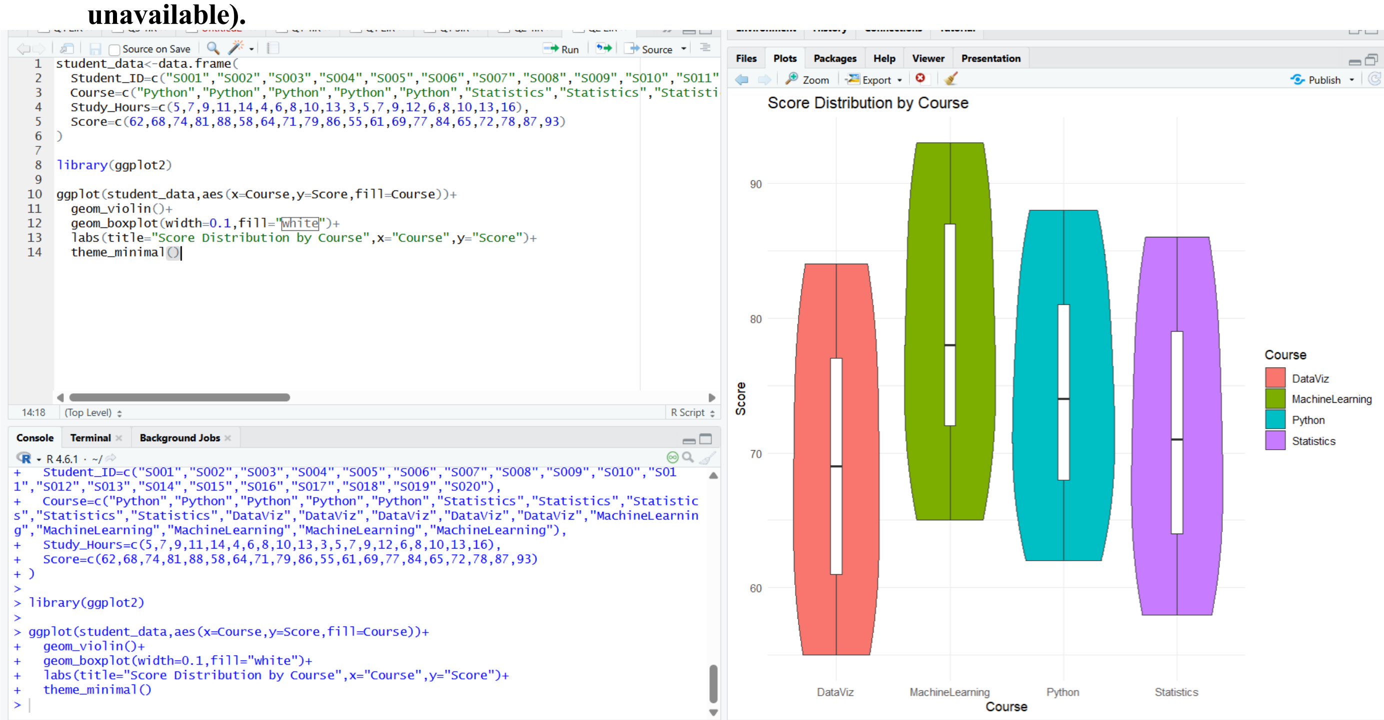
Dataset: Student_Performance.csv

Assessment Task

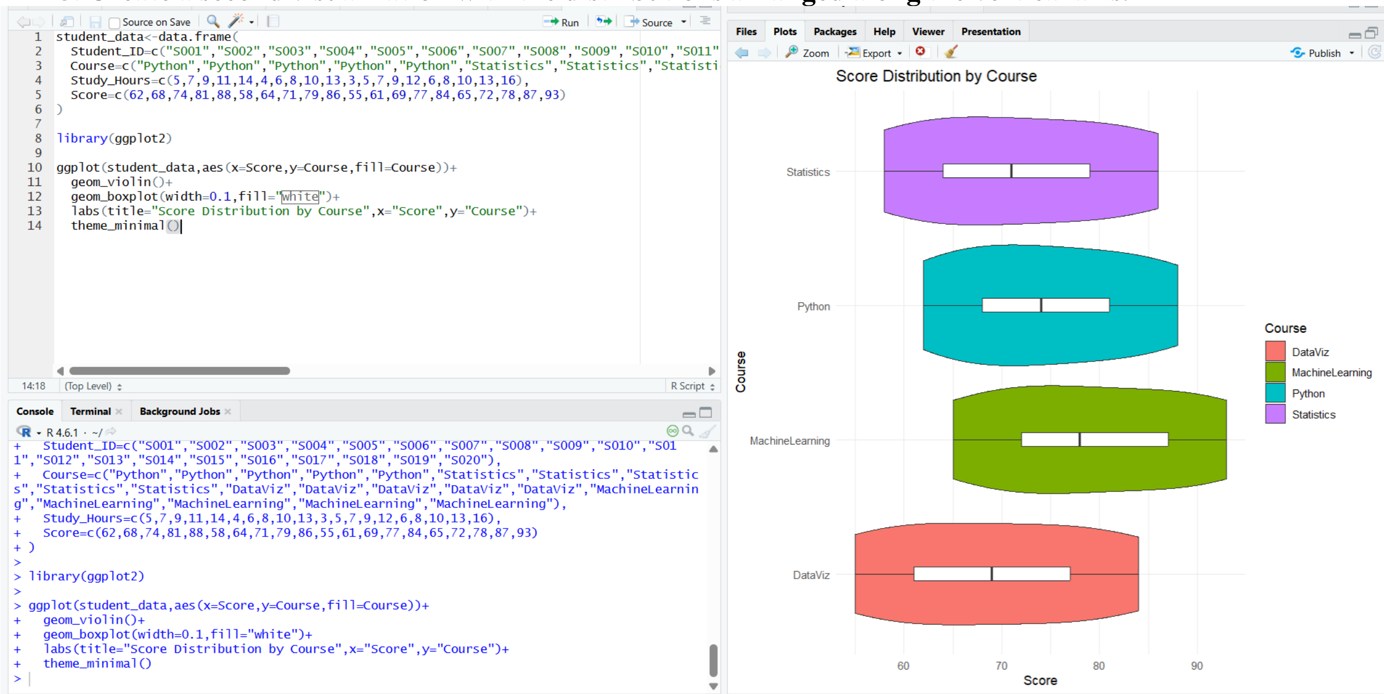
Using the Student_Performance dataset, investigate how scores are distributed across courses.

1. Visualize the score distribution for all four courses at the same time.

2. Create bean plots (or an equivalent violin/density-based visualization if a bean-plot package is unavailable).



3. Create a second visualization with the distributions arranged along the vertical axis.



4. Compare the center, spread, and shape of the distributions and determine which course has the highest and most consistent performance.

MachineLearning has the highest overall performance, with scores ranging from **65 to 93** and the highest average score among the four courses. **DataViz** has the lowest average score. The distributions show a similar increasing pattern, but MachineLearning has a wider spread, while **Python** has the most consistent performance because its scores have a relatively smaller spread. Overall, MachineLearning has the highest performance, while Python is the most consistent.

Dataset:

Student_ID	**Course**	**Study_Hours**	**Score**
S001	Python	5	62
S002	Python	7	68
S003	Python	9	74
S004	Python	11	81
S005	Python	14	88
S006	Statistics	4	58
S007	Statistics	6	64
S008	Statistics	8	71
S009	Statistics	10	79
S010	Statistics	13	86
S011	DataViz	3	55
S012	DataViz	5	61
S013	DataViz	7	69
S014	DataViz	9	77
S015	DataViz	12	84
S016	MachineLearning	6	65
S017	MachineLearning	8	72
S018	MachineLearning	10	78
S019	MachineLearning	13	87
S020	MachineLearning	16	93

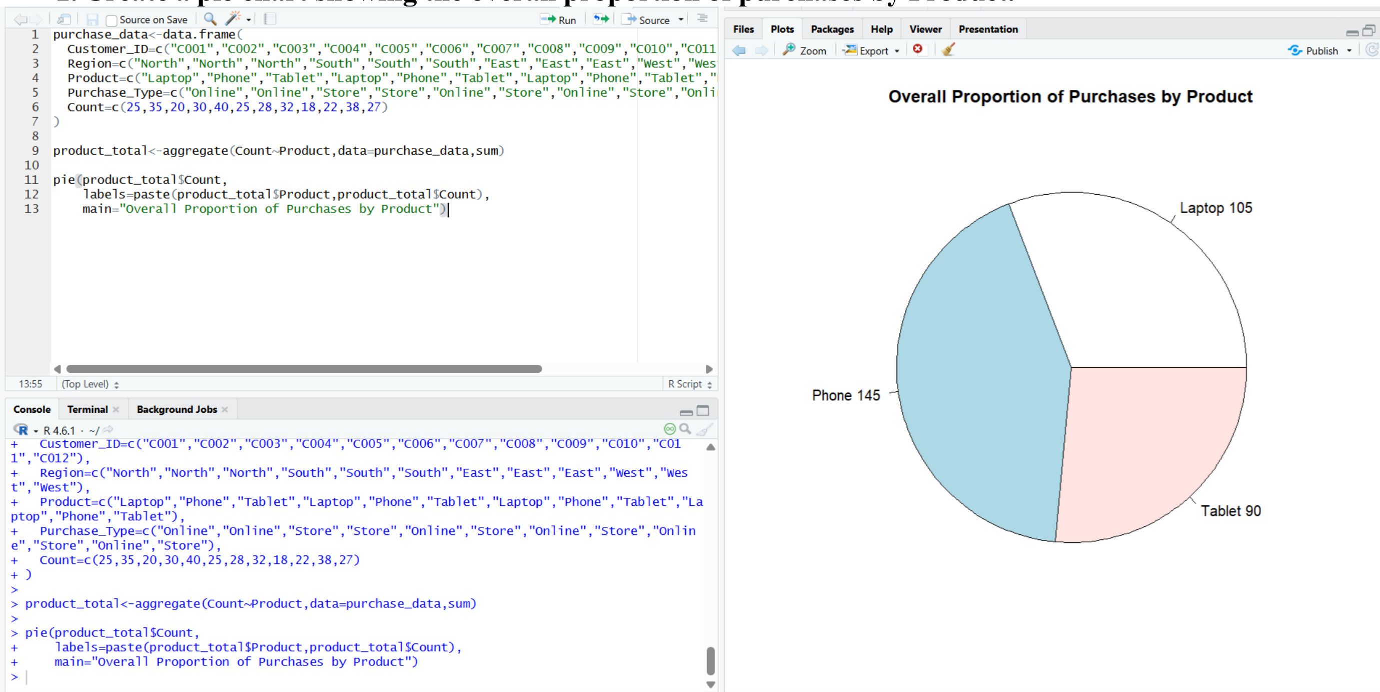
Question 3 – Comparing Proportions with Pie, Side-by-Side and Stacked Bars

Dataset: Customer_Purchases.csv

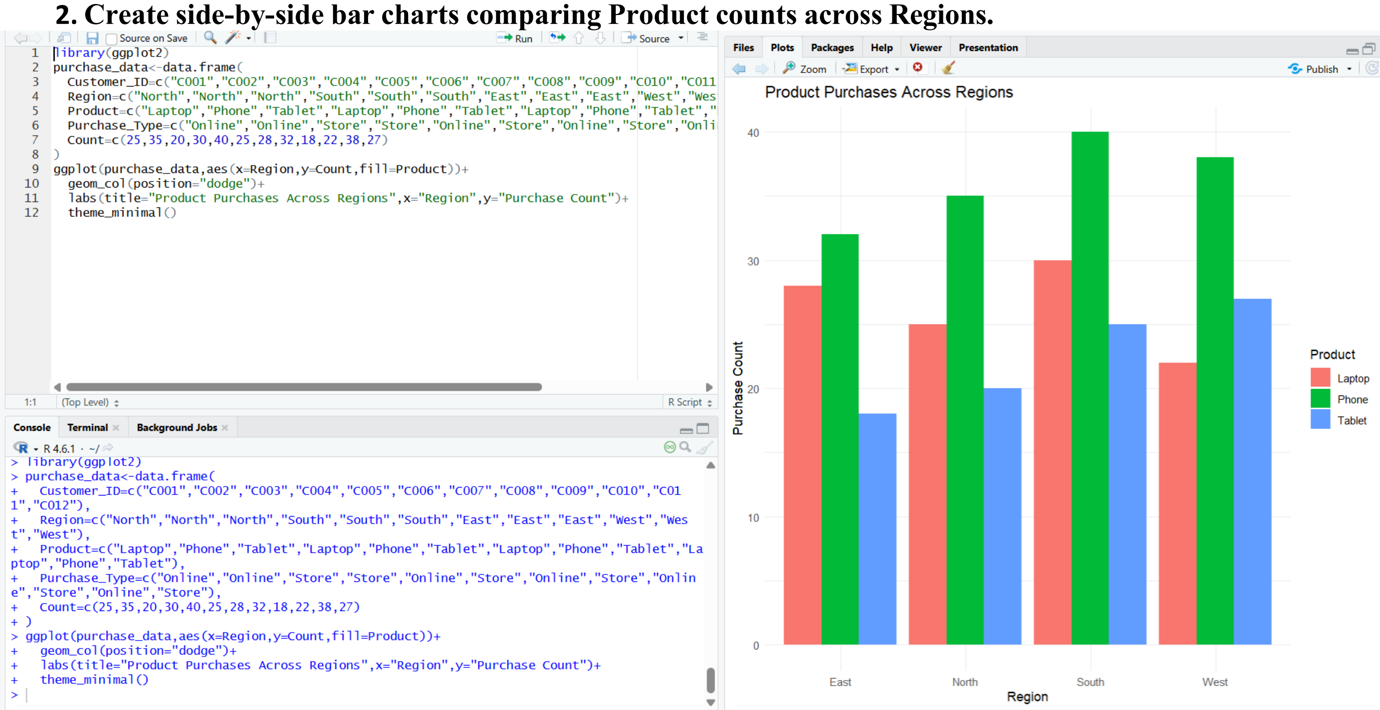
Assessment Task

Using the Customer_Purchases dataset, analyze purchase proportions.

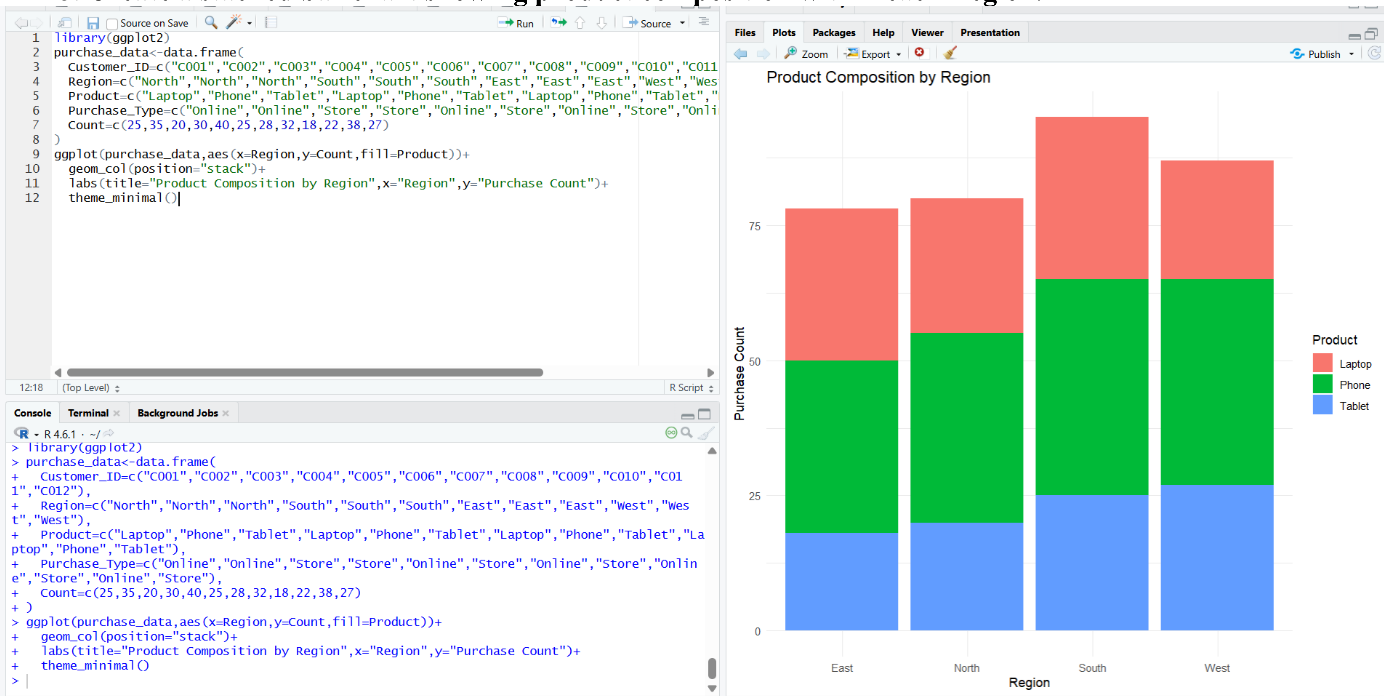
1. Create a pie chart showing the overall proportion of purchases by Product.



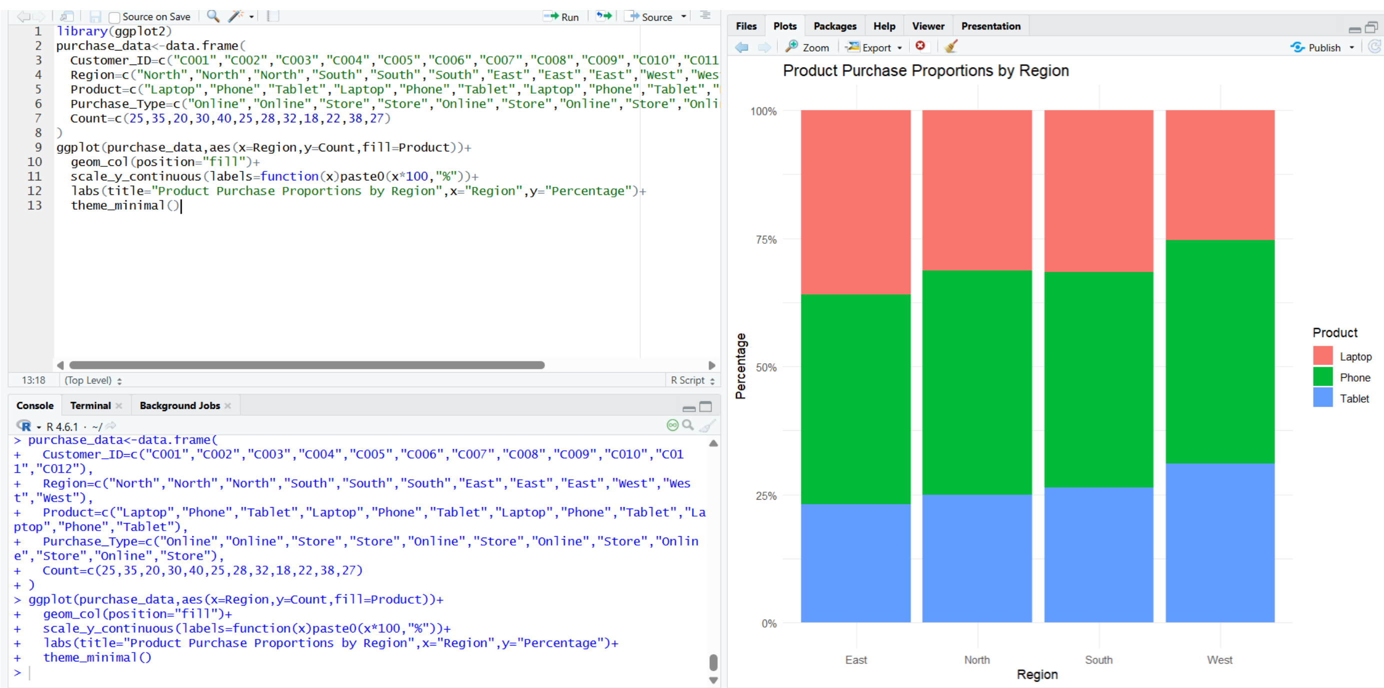
2. Create side-by-side bar charts comparing Product counts across Regions.



3. Create a stacked bar chart showing product composition within each Region.



4. Convert the stacked chart into a 100% stacked bar chart and compare proportions rather than raw counts.



5. Explain why one of these visualizations may be more appropriate than another for comparing regional proportions.

The 100% stacked bar chart is more appropriate for comparing regional proportions because every region has the same total length of 100%. This makes it easier to compare the relative proportion of Laptop, Phone, and Tablet purchases across regions. A regular stacked bar chart shows raw counts, so differences in the total number of purchases can make proportional comparisons more difficult.

Dataset:

Customer_ID	Region	Product	Purchase_Type	Count
C001	North	Laptop	Online	25
C002	North	Phone	Online	35
C003	North	Tablet	Store	20
C004	South	Laptop	Store	30
C005	South	Phone	Online	40
C006	South	Tablet	Store	25
C007	East	Laptop	Online	28
C008	East	Phone	Store	32
C009	East	Tablet	Online	18
C010	West	Laptop	Store	22
C011	West	Phone	Online	38
C012	West	Tablet	Store	27

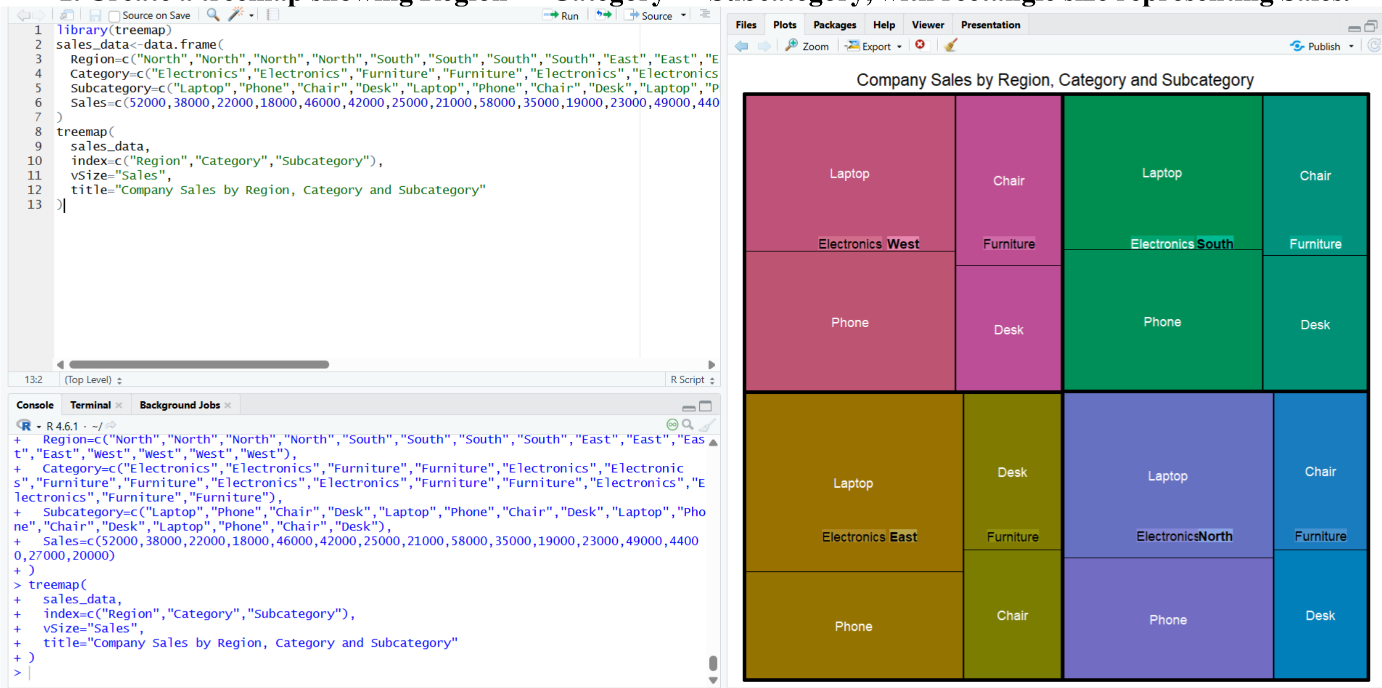
Question 4 – Nested Proportions Using Treemap and Sunburst Charts

Dataset: Company_Sales.csv

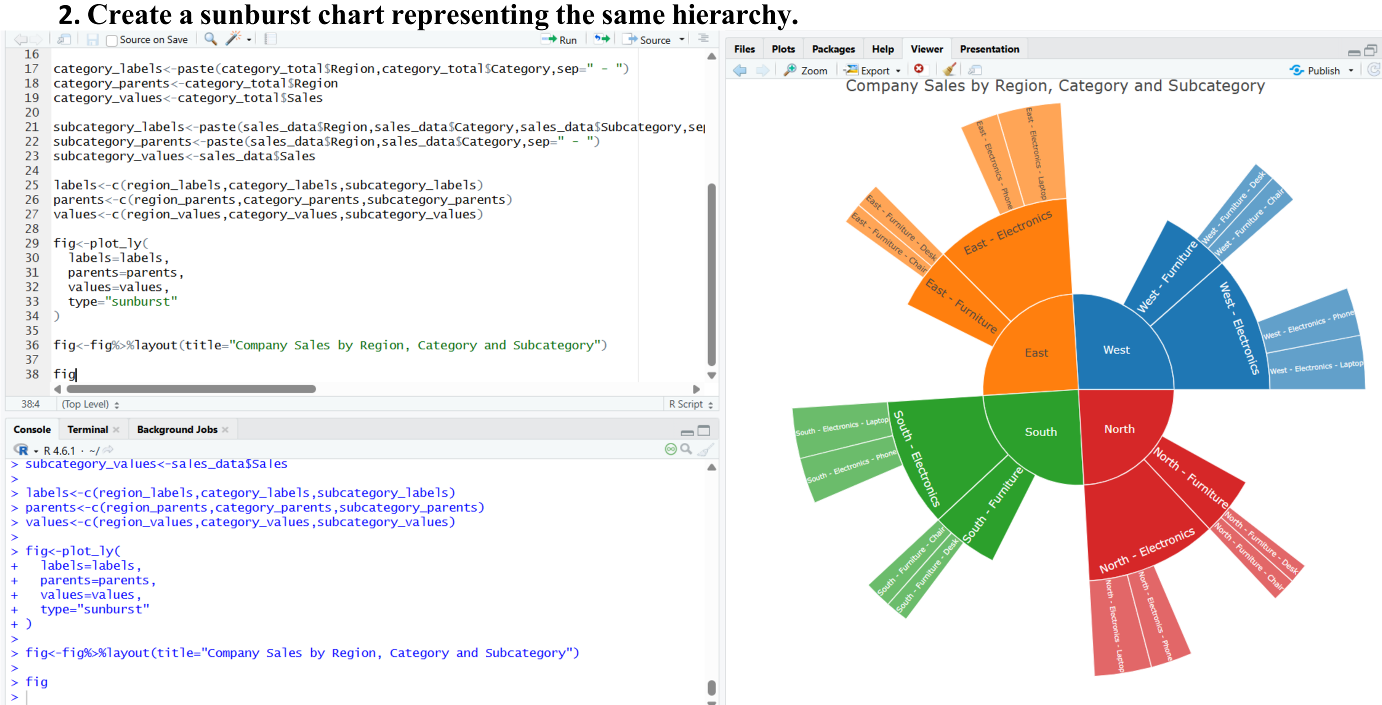
Assessment Task

Using the Company_Sales dataset, visualize nested proportions in the sales hierarchy.

1. Create a treemap showing Region → Category → Subcategory, with rectangle size representing Sales.



2. Create a sunburst chart representing the same hierarchy.



3. Identify the largest region, category, and subcategory by sales.

1. Largest Region

Calculate total sales for each region:

- **North:** 52,000 + 38,000 + 22,000 + 18,000 = **130,000**
- **South:** 46,000 + 42,000 + 25,000 + 21,000 = **134,000**
- **East:** 58,000 + 35,000 + 19,000 + 23,000 = **135,000**
- **West:** 49,000 + 44,000 + 27,000 + 20,000 = **140,000**

Largest Region: West — $140,000

2. Largest Category
Across all regions:
Electronics:
52,000 + 38,000 + 46,000 + 42,000 + 58,000 + 35,000 + 49,000 + 44,000
= **$364,000**
Furniture:
22,000 + 18,000 + 25,000 + 21,000 + 19,000 + 23,000 + 27,000 + 20,000
= **$175,000**
Largest Category: Electronics — $364,000
3. Largest Subcategory

- Laptop = 52,000 + 46,000 + 58,000 + 49,000 = **$205,000**
- Phone = 38,000 + 42,000 + 35,000 + 44,000 = **$159,000**
- Chair = 22,000 + 25,000 + 19,000 + 27,000 = **$93,000**
- Desk = 18,000 + 21,000 + 23,000 + 20,000 = **$82,000**

Largest Subcategory: Laptop — $205,000

4. Compare the readability of the treemap and sunburst chart for identifying nested proportions.

The treemap is easier for comparing the relative sales sizes because the area of each rectangle directly represents sales, making large and small values easier to compare. The sunburst chart is better for understanding the hierarchical relationship between Region, Category, and Subcategory. Therefore, the treemap is more suitable for comparing sales magnitude, while the sunburst is more suitable for understanding the nested structure.

5. State one potential limitation of each visualization.

- **Treemap limitation:** It can become difficult to read when there are many small rectangles or when categories have very similar sizes, making precise comparisons difficult.
- **Sunburst limitation:** It can become difficult to interpret when there are many hierarchical levels or small segments, as the outer rings become crowded and harder to compare.

Dataset:

Region	Category	Subcategory	Sales
North	Electronics	Laptop	52000
North	Electronics	Phone	38000
North	Furniture	Chair	22000
North	Furniture	Desk	18000
South	Electronics	Laptop	46000
South	Electronics	Phone	42000
South	Furniture	Chair	25000
South	Furniture	Desk	21000
East	Electronics	Laptop	58000
East	Electronics	Phone	35000
East	Furniture	Chair	19000
East	Furniture	Desk	23000
West	Electronics	Laptop	49000
West	Electronics	Phone	44000
West	Furniture	Chair	27000
West	Furniture	Desk	20000

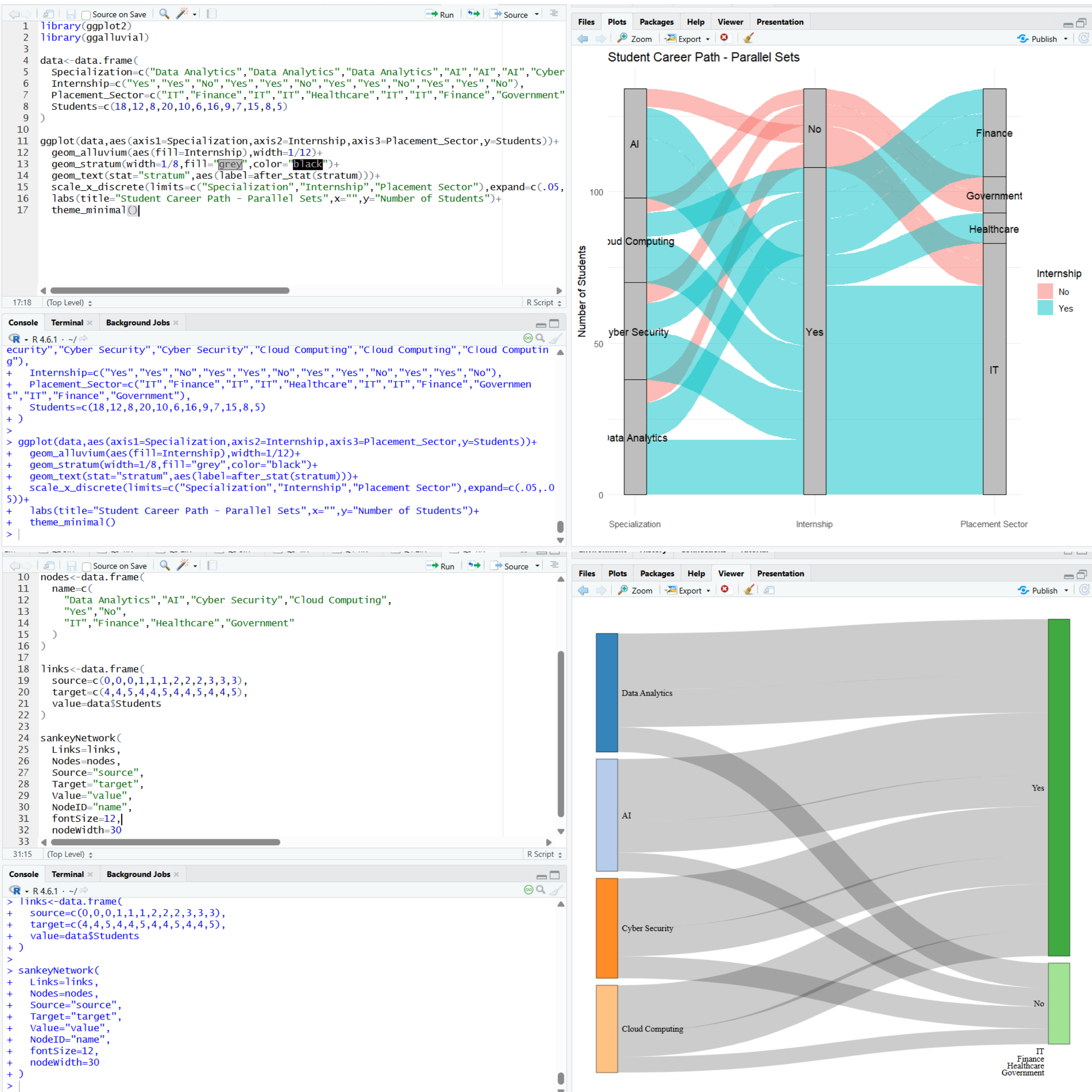
Question 5 – Flow-Based Proportions with Sankey Diagram and Parallel Sets

Dataset: Student_Career_Path.csv

Assessment Task

Using the Student_Career_Path dataset, analyze how students flow from specialization to internship status and placement sector.

1. Create a Sankey diagram showing Specialization → Internship → Placement_Sector, with flow width based on Students.



2. Create a parallel sets visualization for the same categorical relationships.

3. Identify the major flows and compare the placement patterns of students with and without internships.
The major flow is AI → Internship Yes → IT, with 20 students, followed by Data Analytics → Internship Yes → IT with 18 students and Cyber Security → Internship Yes → IT with 16 students. Students with internships show a strong tendency toward the IT sector, while students without internships are mainly placed in IT and Government sectors. Overall, the data shows that students with internships have more diverse placement opportunities, with IT being the dominant sector.

4. Explain which visualization communicates the flow structure more clearly and why.

The Sankey diagram communicates the flow structure more clearly because the width of each flow directly represents the number of students moving from specialization to internship status and then to placement sector. It makes the major and dominant flows easy to identify. The parallel sets visualization is useful for understanding the relationships between the categorical variables, but the flow magnitudes are generally easier to interpret in the Sankey diagram.

5. Discuss one way the chart could become misleading if proportions are represented using inappropriate scales, colors, or labels.

The visualization can become misleading if the flow widths are not proportional to the actual number of students. Using inappropriate colors or labels can also make some flows appear more important than they really are. Therefore, the scale, colors, and labels should accurately represent the student counts and clearly distinguish the different categories.

Dataset:

Student_Group	Specialization	Internship	Placement_Sector	Students
G1	Data Analytics	Yes	IT	18
G2	Data Analytics	Yes	Finance	12
G3	Data Analytics	No	IT	8
G4	AI	Yes	IT	20
G5	AI	Yes	Healthcare	10
G6	AI	No	IT	6
G7	Cyber Security	Yes	IT	16
G8	Cyber Security	Yes	Finance	9
G9	Cyber Security	No	Government	7
G10	Cloud Computing	Yes	IT	15
G11	Cloud Computing	Yes	Finance	8
G12	Cloud Computing	No	Government	5

Expected Student Deliverables

- R script (.R) containing data creation/import, data preparation, analysis, and visualization code.
- All required plots with meaningful titles, axis labels, legends, and appropriate scales.
- A short interpretation for each visualization.
- A brief comparison of alternative visualization methods where requested.
- A concluding statement identifying important patterns, differences, or potential misleading visual encodings.

Suggested Assessment Criteria

Criterion	Weight
Dataset preparation	20%
Correct R programming and use of visualization packages	25%
Appropriate choice and construction of visualizations	25%

Interpretation and analytical reasoning	20%

Clarity, labeling, and presentation	10%